

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/ CE/EE/IT)

SEM - II University Examination, 2011-12

MA - 102 Engineering Mathematics-II

Date: 07.05.12, Monday

Time: 10.00 a. m to 01.00 p. m.

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks.

SECTION-I

Q-1 (a) Solve by the method of variation of parameters : $\frac{d^2y}{dx^2} + 49y = \sec 7x$. [05]

(b) The equation $M dx + N dy = 0$ is exact if _____. [01]

(c) Find the differential equation corresponding to $y(x) = ae^{3x}$. [01]

Q-2 (a) Solve $\frac{dy}{dx} + 2y \tan x = \sin x$; $y = 0$ for $x = \frac{\pi}{3}$. [03]

(b) Solve the initial-value problem $y''' - 6y'' + 11y' - 6y = 0$; $y(0) = y'(0) = 0, y''(0) = 2$. [04]

(c) Solve $x^3 y''' + 2x^2 y'' + 2y = 10(x + \frac{1}{x})$. [04]

(d) Solve $(x^2 y - 2xy^2) dx - (x^3 - 3x^2 y) dy = 0$. [03]

OR

Q-2 (a) Solve $\sec^2 y \frac{dy}{dx} + x \tan y = x^3$. [03]

(b) Solve $y''' + 3y'' + 3y' + y = 8e^x + x + 3$. [04]

(c) Solve $y'' + 3y' = 28 \cosh 4x$. [03]

(d) Solve $\frac{dx}{dt} = 7x - y, \frac{dy}{dt} = 2x + 5y$. [04]

Q-3 (a) Solve $z^2(p^2 + q^2 + 1) = 1$. [03]

(b) Solve $(mz - ny) \frac{\partial z}{\partial x} + (nx - lz) \frac{\partial z}{\partial y} = ly - mx$. [04]

(c) From the Partial differential equation (i) $z = (x+a)(y+b)$ (ii) $z = f(x^2 + y^2)$. [04]

(d) Solve (i) $p^2 - q^2 = 1$ (iii) $z - px = qy - 10\sqrt{pq}$. [03]

OR

Q-3 (a) Solve $x(z^2 - y^2) \frac{\partial z}{\partial x} + y(x^2 - z^2) \frac{\partial z}{\partial y} = z(y^2 - x^2)$. [04]

(b) Solve $p^2(1 + q^2) = q(z - a)$. [03]

(c) Solve (i) $p + q + pq = 0$ (ii) $p - x^2 = q + y^2$. [04]

(d) (i) The solution of $xp + yq = z$ is _____. [03]

(ii) The solution of $p + q = z$ is _____.

SECTION-II

Q-4 (a) Change the order of integration in the integral $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx$ and evaluate it. [04]

(b) Prove $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$. [03]

Q-5 (a) Find the eigenvalues and corresponding eigenvectors for the [05]

$$\text{matrix } A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

(b) Express $f(x) = 4x^3 - 3x^2 + 2x + 1$ in terms of Legendre polynomials. [03]

(c) Evaluate (i) $\int_0^{\infty} e^{-x^3} dx$. [02]

(d) Find the area between the parabola $y^2 = 4ax$ and $x^2 = 4ay$. [04]

OR

Q-5 (a) State Cayley Hamilton theorem. Find the characteristic equation of the [05]

$$\text{matrix } A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \text{ and hence evaluate the matrix equation}$$

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I.$$

(b) Prove $P(1) = 1$. [02]

(c) Evaluate $\iint r^3 \sin 2\theta \, dr \, d\theta$ over the area bounded in the first quadrant R between the circles $r=2$ and $r=4$. [04]

(d) Show that Evaluate $\int_0^1 x^m \left(\log \frac{1}{x} \right) dx$. [03]

Q-6 (a) If $B = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, show that B is Hermitian matrix and iB is a Skew-Hermitian matrix. [05]

(b) Evaluate $\int_0^n x^n (n-x)^p \, dx$. [02]

(c) Evaluate $\iiint_V z(x^2 + y^2) \, dx$ over the cylinder $x^2 + y^2 = 1$ intercepted by the plane $z=2$ and $z=3$. [04]

(d) Prove that $\frac{d}{dx} [x^{n+1} J_{n+1}(x)] = x^{n+1} J_n(x)$. [03]

OR

Q-6 (a) Evaluate $\iint_R dA$; R is the region bounded by $x=0, x=1, y=x^2, y=2-x$. [04]

(b) Prove $\operatorname{erfc}(x) + \operatorname{erfc}(-x) = 2$. [03]

(c) Find the volume common to the cylinder $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$. [04]

(d)(i) $\int_{-1}^1 P_3(x) P_4(x) \, dx = \underline{\hspace{2cm}}$. (ii) $\int_{-1}^1 P_6^2(x) \, dx = \underline{\hspace{2cm}}$. [03]